D-CAPS 2027 Call for Papers

International Workshop on Data, Computing, and Communication Infrastructure for Autonomous Physical Systems

April 4, 2027 · Cal Poly Pomona, Pomona, California, USA

Proposed workshop at the Fifth IEEE International Conference on Mobility: Operations, Services, and Technologies (IEEE MOST 2027)

Proposed paper submission deadline: February 12, 2027 · Submit through D-CAPS HotCRP

Workshop website: int-lab.github.io/dcaps2027

Advances in sensing, embedded computing, and control are expanding the range of tasks that autonomous physical systems can perform across transportation, manufacturing, and critical infrastructure. Realizing this potential depends on the data, computing, and communication infrastructure that determines which physical states a system can observe, which decisions it can compute, and how reliably it can translate those decisions into action.

The dynamics of a physical task couple the fidelity and freshness of observations with the computational throughput and communication latency required to act on them within a platform's resource and energy budgets. D-CAPS brings together researchers in data systems, computer architecture, and networking to examine how infrastructure design choices shape the performance, dependability, and operating limits of autonomous physical systems. Co-location with IEEE MOST gives participants a forum to examine which design principles generalize across land, air, and maritime mobility and which depend on a platform's physical dynamics and operating conditions.

We invite original papers from academia, industry, and government on the design, integration, and evaluation of infrastructure within a single platform or across multiple systems, as well as general-purpose architectures and services with potential applications to autonomous physical systems. We welcome foundational research, early results, prototype systems, measurement studies, and lessons from deployment.

Topics of interest

- **Data infrastructure** — Systems for acquiring, storing, managing, and processing data.
- **Computing infrastructure** — Hardware and software architectures for onboard, edge, and cloud computing.
- **Communication infrastructure** — Architectures and protocols for exchanging information within and across systems.
- **Sensing and actuation infrastructure** — Hardware and software that enable autonomous physical systems to observe and act on their environment.
- **AI and machine learning infrastructure** — Systems for training, deploying, and serving models.
- **Systems architecture and integration** — Design and integration of sensing, computing, communication, and control.
- **Infrastructure management and orchestration** — Methods for provisioning, configuring, and managing data, computing, and communication resources for autonomous physical systems.
- **Interoperability and standards** — Interfaces and standards that allow components from different vendors or platforms to work together.
- **Dependability, security, and privacy** — Methods for maintaining reliable operation and protecting systems and data.
- **Testbeds, measurement, and deployment** — Experimental platforms, evaluation methods, and lessons from real-world deployments.

Important dates

Milestone	Date
Paper submission (proposed)	February 12, 2027
Author notification	February 28, 2027
Camera-ready manuscript	To be confirmed
In-person workshop	April 4, 2027

How to submit

Submit a PDF of up to **six pages** in the **double-column IEEE conference format**. The limit includes references, figures, tables, acknowledgments, and appendices. Number all pages, embed the fonts, and check that illustrations are legible in grayscale. IEEE Word and LaTeX templates

Single-blind review. Include author names and affiliations in the manuscript. Reviewers will see the author list, and their identities will remain confidential.

Submit original, unpublished work that is not under review at another archival venue. With the approval of all authors, enter the title, abstract, complete author list, affiliations, and conflicts of interest in the submission system. Direct requests for post-deadline changes to the workshop co-chairs. D-CAPS submission site

Review and publication

Each paper will receive at least three independent reviews. The committee will consider originality, relevance, technical soundness, clarity, and whether the evidence supports the claims. We welcome analyses, simulations, hardware experiments, and field studies suited to the research question.

At least one author of each accepted paper must register and present in person. Authors must submit the camera-ready PDF and complete the IEEE publication forms. Accepted and presented papers will appear in the IEEE MOST 2027 proceedings and will be submitted to IEEE Xplore.

Workshop co-chairs

- **Qiang Liu**, University of Nebraska–Lincoln · qiang.liu@unl.edu
- **Nakjung Choi**, Nokia Bell Labs · nakjung.choi@nokia-bell-labs.com

Steering committee

- **Weisong Shi**, University of Delaware
- **Onur Altintas**, Head of InfoTech Labs, Toyota North America R&D

TPC members

- **Zheng Dong**, Wayne State University
- **Qi Chen**, Toyota Motor North America
- **Ahan Kak**, Nokia Bell Labs
- **Haoxin Wang**, Georgia State University

IEEE MOST 2027 workshop information